

Unit 1

Cloud Services and Cloud Models

Introduction to Cloud

Cloud Computing

Cloud computing delivers services over the internet using remote servers, offering scalability and accessibility. Traditional computing relies on local servers, requiring upfront hardware investments and maintenance. The key distinction lies in where and how services are hosted, impacting flexibility, cost-effectiveness, and ease of access.

Storing and accessing of data and computing services over the internet.

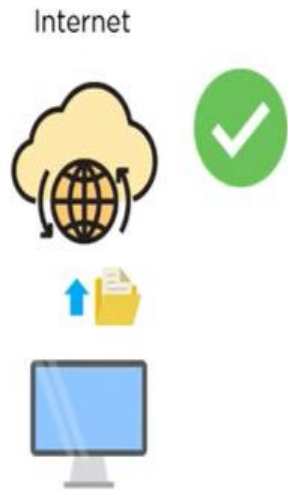
Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale.

What is Cloud Computing?

Rather than managing files on a local storage device, cloud computing makes it possible to save them over internet



Storage device

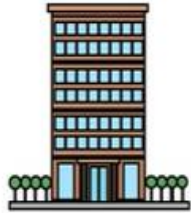


Save files

Why Cloud Computing?

On-premise vs Cloud Computing

ON-PREMISE



- Higher pay, less scalability
- Allot huge space for servers
- Appoint a team for hardware and software maintenance
- Poor data security
- Less chance of data recovery

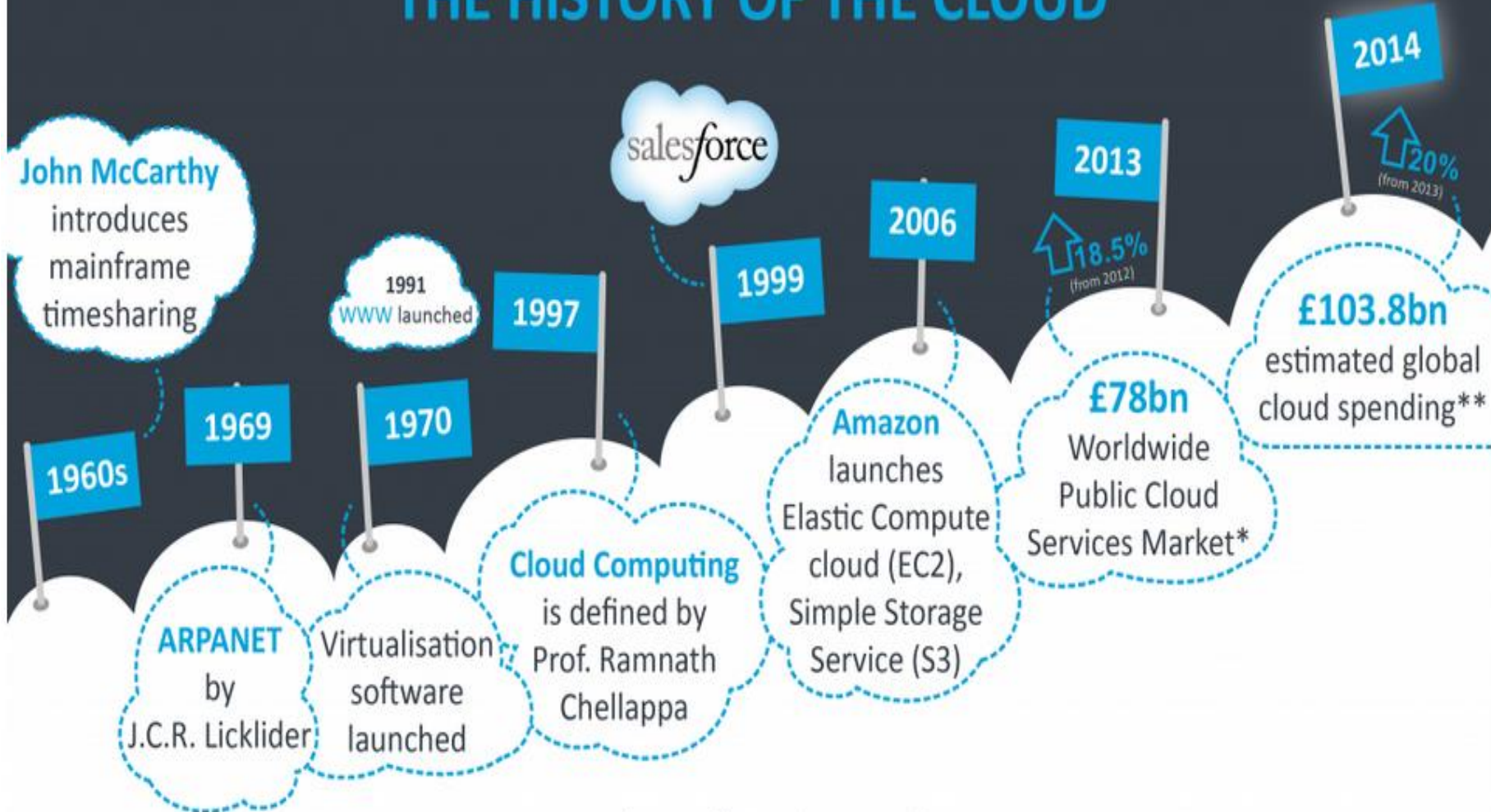


- Pay for what you use
 - Scale up= pay more
 - Scale down= pay less
- No server space required
- No experts required for hardware and software maintenance
- Better data security
- Disaster recovery

Aspect	Cloud Computing	Traditional Computing
Definition	Cloud Computing refers to delivery of different services such as data and programs through internet on different servers.	Traditional Computing refers to delivery of different services on local server.
Infrastructure Location	Cloud Computing takes place on third-party servers that is hosted by third-party hosting companies.	Traditional Computing takes place on physical hard drives and website servers.
Data Accessibility	Cloud Computing is ability to access data anywhere at any time by user.	User can access data only on system in which data is stored.
Cost Effectiveness	Cloud Computing is more cost effective as compared to tradition computing as operation and maintenance of server is shared among several parties that in turn reduce cost of public services.	Traditional Computing is less cost effective as compared to cloud computing because one has to buy expensive equipment's to operate and maintain server.
User-Friendliness	Cloud Computing is more user-friendly as compared to traditional computing because user can have access to data anytime anywhere using internet.	Traditional Computing is less user-friendly as compared to cloud computing because data cannot be accessed anywhere and if user has to access data in another system, then he need to save it in external storage medium.

Aspect	Cloud Computing	Traditional Computing
Internet Dependency	Cloud Computing requires fast, reliable and stable internet connection to access information anywhere at any time.	Traditional Computing does not require any internet connection to access data or information.
Storage and Computing Power	Cloud Computing provides more storage space and servers as well as more computing power so that applications and software run must faster and effectively.	Traditional Computing provides less storage as compared to cloud computing.
Scalability and Elasticity	Cloud Computing also provides scalability and elasticity i.e., one can increase or decrease storage capacity, server resources, etc., according to business needs.	Traditional Computing does not provide any scalability and elasticity.
Maintenance and Support	Cloud service is served by provider's support team.	Traditional Computing requires own team to maintain and monitor system that will need a lot of time and efforts.
Software Delivery Model	Software is offered as an on-demand service (SaaS) that can be accessed through subscription service.	Software is purchased individually for every user and requires to be updated periodically.

THE HISTORY OF THE CLOUD

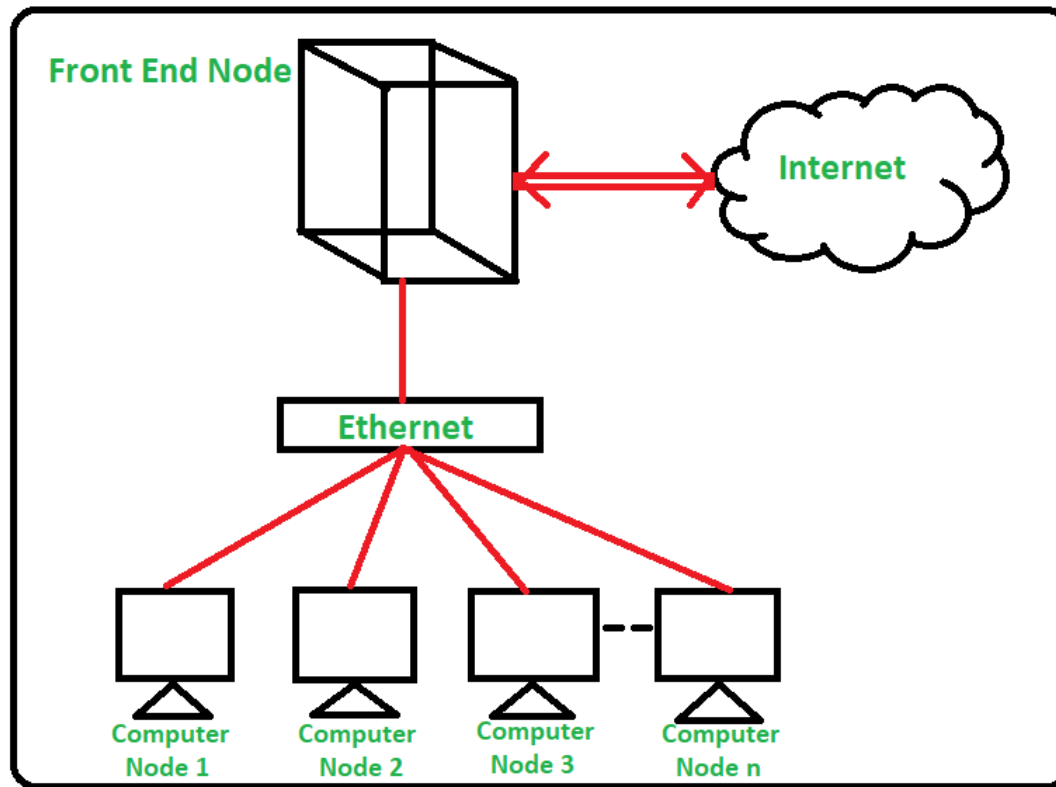


* Gartner, ** Constellation Research

Cluster Computing

- ▶ Cluster computing refers to the process of sharing the computation task to multiple computers of the cluster. The number of computers are connected on a network and they perform a single task by forming a Cluster of computers where the process of computing is called as cluster computing.
- ▶ Cluster Computing is a high performance computing framework which helps in solving more complex operations more efficiently with a faster processing speed and better data integrity. Cluster Computing is a networking technology that performs its operations based on the principle of distributed systems.

Cluster Computing



Difference between Cloud Computing and Cluster Computing

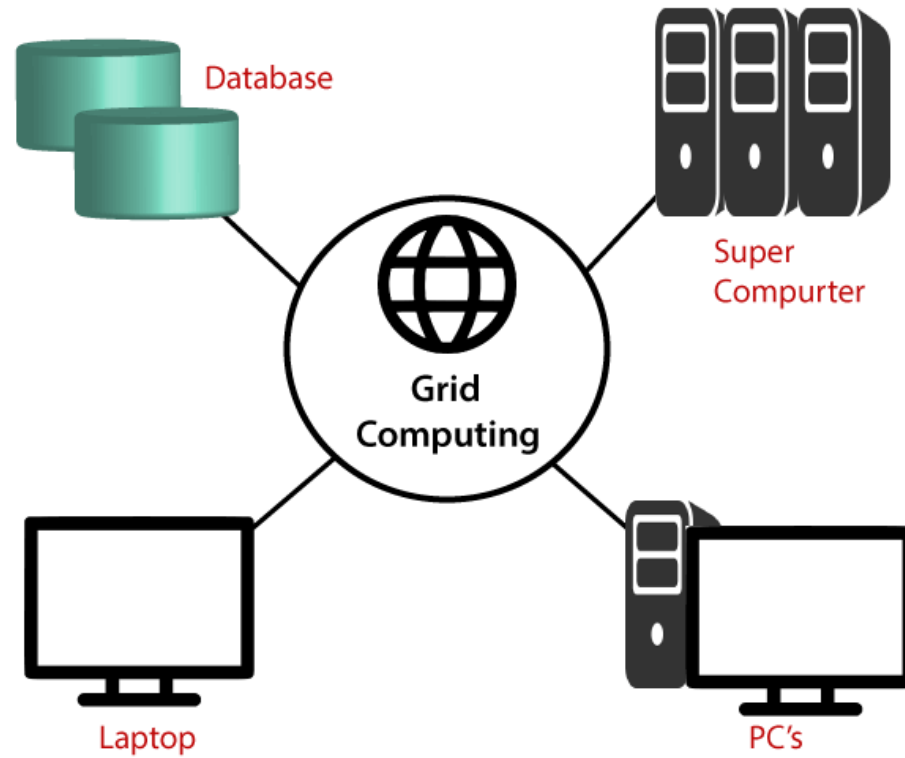
Serial Number	Category	Cloud Computing	Cluster Computing
1.	Goal	Providing on demand IT resources and services.	Performing a complex task in a modular approach.
2.	Resource Sharing	Specific assigned resources are not shareable.	Specific assigned resources are not shareable.
3.	Resource type	In cloud computing there is heterogeneous resource type.	In Cluster Computing there is homogeneous resource type.
4.	Virtualization	Virtualization hardware and software resources.	No virtualization resources.
5.	Security	Security through isolation can be achieved.	Security through node credential can be achieved.
6.	Initial Cost	Initial capital cost for setup is very low.	Initial capital cost for setup is very high.

7.	Security Requirement	Very low	Very high
8.	Maintenance	Requires low maintenance.	Requires little more maintenance.
9.	Hardware	No hardware requirement physically.	More hardware requirement physically.
10.	Node OS	Multiple OS runs in VM	Windows, Linux
11.	User Management	User management is centralized or decentralized to vendor/third party.	User management is centralized.
12.	Scalability	Allowed	Limited
13.	Architecture	In Cloud Computing User chosen architecture.	In Cluster Computing Cluster oriented architecture

Grid Computing

- ▶ Grid computing is a distributed architecture of multiple computers connected by networks to accomplish a joint task. These tasks are compute-intensive and difficult for a single machine to handle. Several machines on a network collaborate under a common protocol and work as a single virtual supercomputer to get complex tasks done. This offers powerful virtualization by creating a single system image that grants users and applications seamless access to IT capabilities.
- ▶ Grid computing operates by running specialized software on every computer involved in the grid network. The software coordinates and manages all the tasks of the grid. Fundamentally, the software segregates the main task into subtasks and assigns the subtasks to each computer. This allows all the computers to work simultaneously on their respective subtasks. Upon completion of the subtasks, the outputs of all computers are aggregated to complete the larger main task.
- ▶ The software allows computers to communicate and share information on the portion of the subtasks being carried out. As a result, the computers can consolidate and deliver a combined output for the assigned main task.

Grid Computing

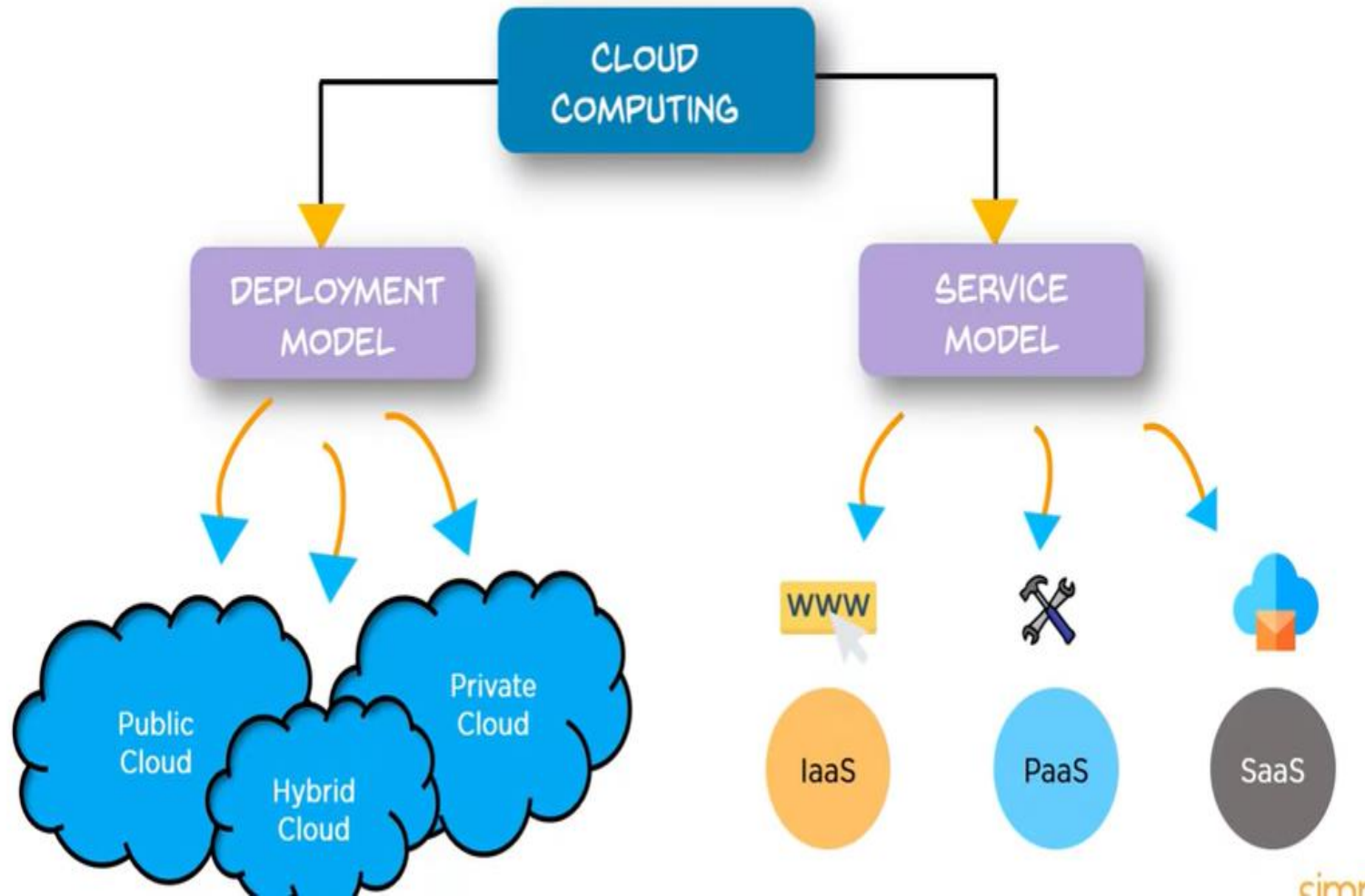


Difference between Cloud Computing and Grid Computing

Cloud Computing	Grid Computing
Cloud computing is a Client-server computing architecture.	While it is a Distributed computing architecture.
Cloud computing is a centralized executive.	While grid computing is a decentralized executive.
In cloud computing, resources are used in centralized pattern.	While in grid computing, resources are used in collaborative pattern.
It is more flexible than grid computing.	While it is less flexible than cloud computing.
In cloud computing, the users pay for the use.	While in grid computing, the users do not pay for use.
Cloud computing is a high accessible service.	While grid computing is a low accessible service.

It is highly scalable as compared to grid computing.	While grid computing is low scalable in comparison to cloud computing.
It can be accessed through standard web protocols.	While it is accessible through grid middleware.
Cloud computing is based on service-oriented.	Grid computing is based on application-oriented.
Cloud computing uses service like IAAS , PAAS, SAAS .	Grid computing uses service like distributed computing, distributed information.

Types of Cloud Computing



Cloud deployment models

When adopting cloud architecture, there are three different types of cloud deployment models that help deliver cloud computing services: public cloud, private cloud, and hybrid cloud.

- ▶ Public cloud

Public clouds deliver resources, such as compute, storage, network, develop-and-deploy environments, and applications over the internet. They are owned and run by third-party cloud service providers like Google Cloud.

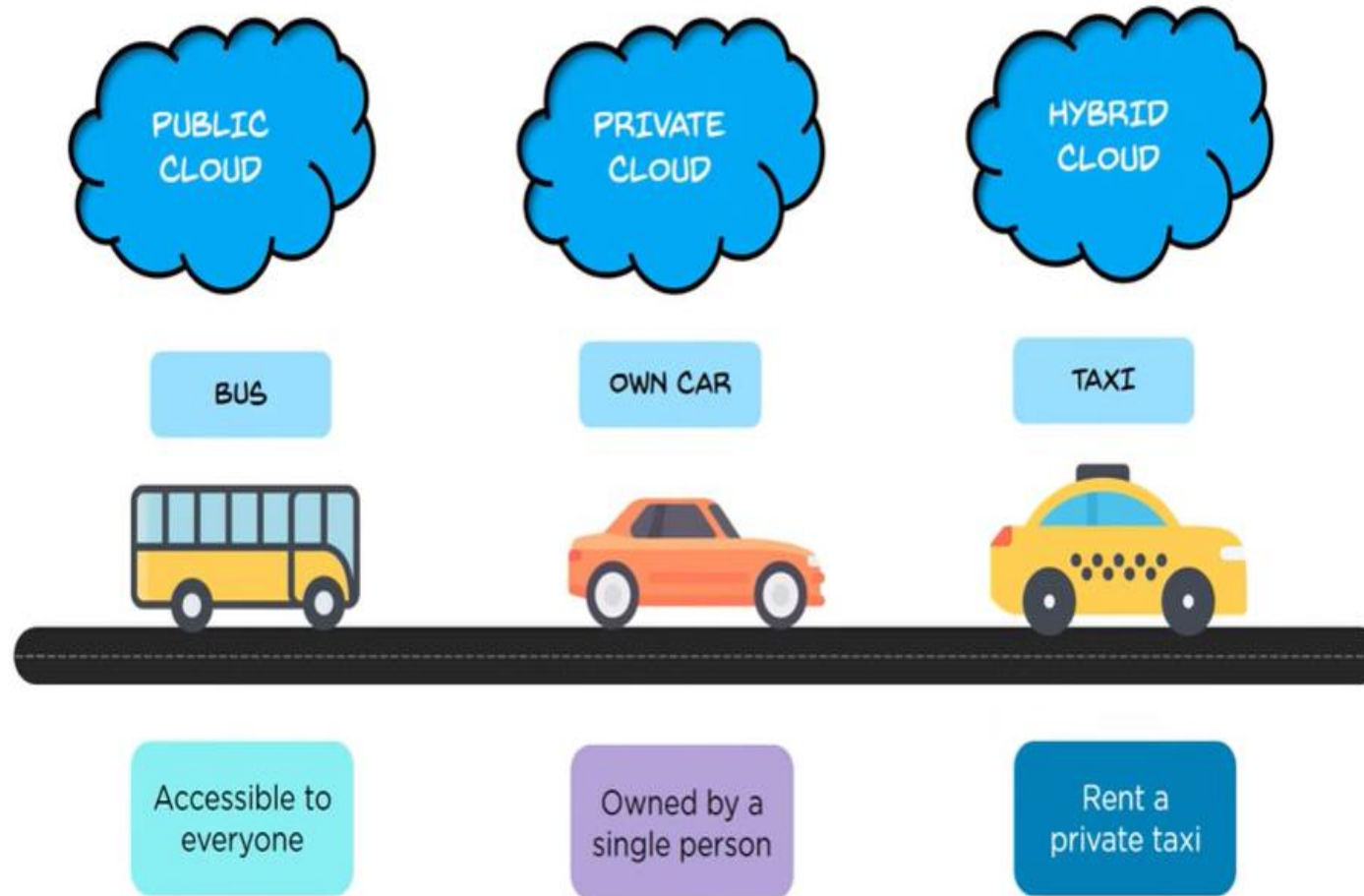
- ▶ Private cloud

Private clouds are built, run, and used by a single organization, typically located on-premises. They provide greater control, customization, and data security but come with similar costs and resource limitations associated with traditional IT environments.

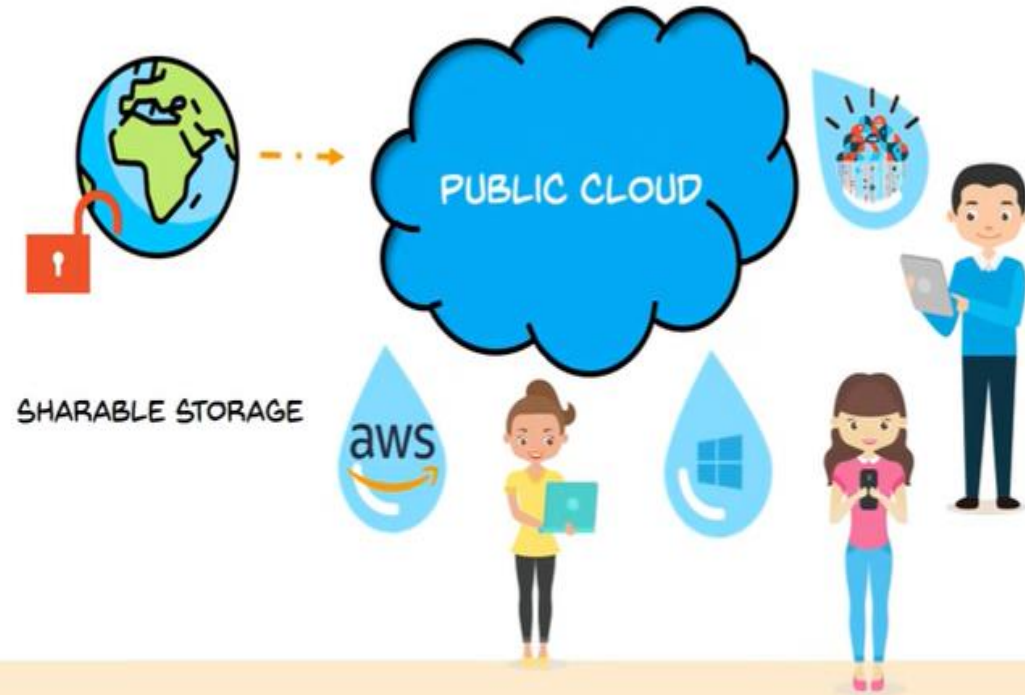
- ▶ Hybrid cloud

Environments that mix at least one private computing environment (traditional IT infrastructure or private cloud, including edge) with one or more public clouds are called hybrid clouds. They allow you to leverage the resources and services from different computing environments and choose which is the most optimal for the workloads.

Types of Deployment Models



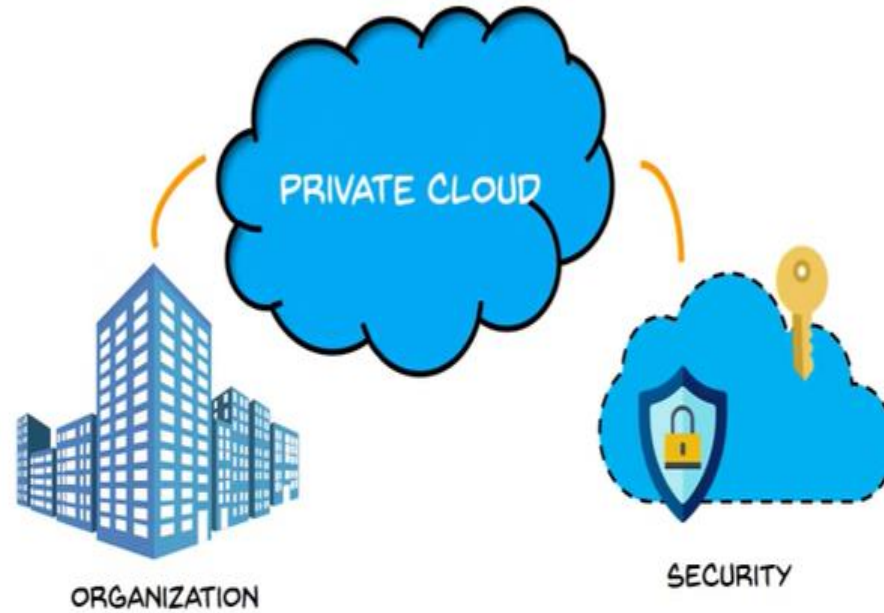
Public Cloud



The cloud infrastructure is made available to the general public over the internet and is owned by a cloud provider

Example: AWS, Microsoft Azure, IBM's Blue Cloud and Sun Cloud

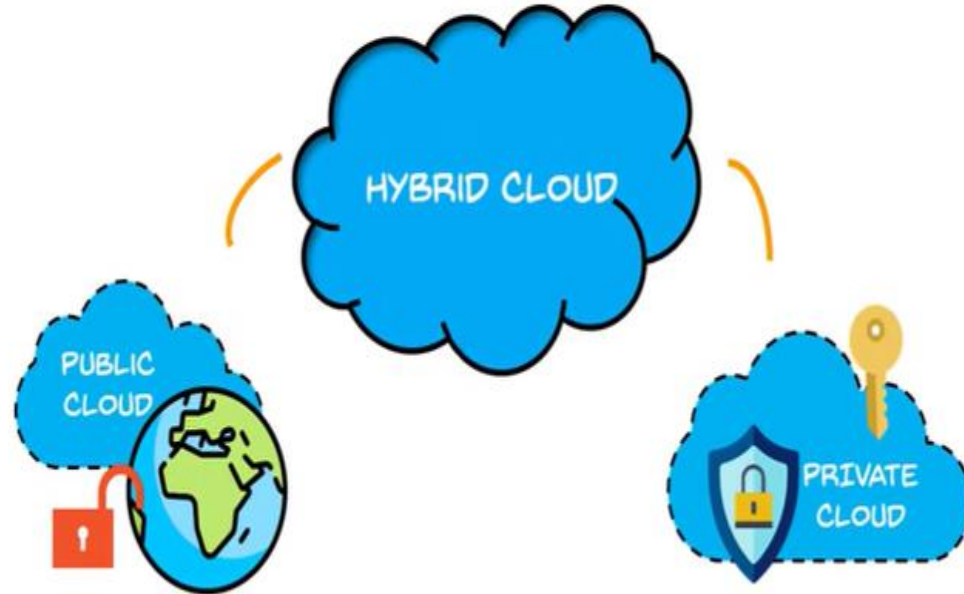
Private Cloud



The cloud infrastructure is exclusively operated by a single organization. It can be managed by the organization or a third party and may exist on-premise or off-premise

Example: AWS, VMware

Hybrid Cloud



It consists the functionalities of both public and private cloud

For example:

Federal agencies opt for private clouds when sensitive information is involved
Also, they use the public cloud to share datasets with general public or other government departments

Types of cloud services

Within the cloud deployment models, there are several types of cloud services, including infrastructure, platforms, and software applications. Cloud service models are not mutually exclusive, and you can choose to use more than one in combination or even all of them at once.

Here are the three main cloud service models:

- Infrastructure as a Service (IaaS)

IaaS delivers on-demand infrastructure resources, such as compute, storage, networking, and virtualization. With IaaS, the service provider owns and operates the infrastructure, but customers will need to purchase and manage software, such as operating systems, middleware, data, and applications.

- Platform as a Service (PaaS)

PaaS delivers and manages hardware and software resources for developing, testing, delivering, and managing cloud applications. Providers typically offer middleware, development tools, and cloud databases within their PaaS offerings.

- Software as a Service (SaaS)

SaaS provides a full application stack as a service that customers can access and use. SaaS solutions often come as ready-to-use applications, which are managed and maintained by the cloud service provider.

Types of Service Models

Which cloud service is suitable for you?

WWW

IAAS

If your business needs a virtual machine, opt for Infrastructure as a Service



PAAS

If your company requires a platform for building software products, pick Platform as a Service



SAAS

If your business doesn't want to maintain any IT equipment, then choose Software as a Service



Infrastructure as a Service (IaaS)

IaaS is also known as **Hardware as a Service (HaaS)**. It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS

- ▶ Resources are available as a service
- ▶ Services are highly scalable
- ▶ Dynamic and flexible
- ▶ GUI and API-based access
- ▶ Automated administrative tasks
- ▶ **Example:** DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.

IaaS



- ✓ IaaS is a cloud service that provides basic computing infrastructure
- ✓ Services are available on **PAY-FOR-WHAT-YOU-USE** model
- ✓ IaaS providers include Amazon Web Services, Microsoft Azure and Google Compute Engine
- ✓ Users: IT Administrators

IAAS PRODUCTS AND SERVICES



Platform as a Service (PaaS)

PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

Characteristics of PaaS

- ▶ Accessible to various users via the same development application.
- ▶ Integrates with web services and databases.
- ▶ Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- ▶ Support multiple languages and frameworks.
- ▶ Provides an ability to **"Auto-scale"**.
- ▶ **Example:** AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.

PaaS



- ✓ PaaS provides cloud platforms and runtime environments for developing, testing, and managing applications
- ✓ It allows software developers to deploy applications without requiring all the related infrastructure
- ✓ Users: Software Developers

PAAS PRODUCTS AND SERVICES



Software as a Service (SaaS)

SaaS is also known as "**on-demand software**". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.

Characteristics of SaaS

- ▶ Managed from a central location
- ▶ Hosted on a remote server
- ▶ Accessible over the internet
- ▶ Users are not responsible for hardware and software updates. Updates are applied automatically.
- ▶ The services are purchased on the pay-as-per-use basis
- ▶ **Example:** BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

SaaS



- ✓ In SaaS, cloud providers host and manage the software application on a pay-as-you-go pricing model
- ✓ All software and hardware are provided and managed by a vendor so you don't have to maintain anything
- ✓ Users: End Customers

SAAS PRODUCTS AND SERVICES



Office 365 Google Apps

Difference between IaaS, PaaS, and SaaS

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software.

Anything as a Service (XaaS)

It is also known as Everything as a Service. Most of the cloud service providers nowadays offer anything as a service that is a compilation of all of the above services including some additional services.

Advantages of XaaS:

- ▶ **Scalability:** XaaS solutions can be easily scaled up or down to meet the changing needs of an organization.
- ▶ **Flexibility:** XaaS solutions can be used to provide a wide range of services, such as storage, databases, networking, and software, which can be customized to meet the specific needs of an organization.
- ▶ **Cost-effectiveness:** XaaS solutions can be more cost-effective than traditional on-premises solutions, as organizations only pay for the services.

Disadvantages of XaaS:

- ▶ **Dependence on the provider:** Users are dependent on the XaaS provider for the availability, scalability, and reliability of the service, which can be a risk if the provider experiences outages or other issues.
- ▶ **Limited flexibility:** XaaS solutions may not be able to accommodate certain types of workloads or applications, which can limit the value of the solution for certain organizations.
- ▶ **Limited integration:** XaaS solutions may not be able to integrate with existing systems and data sources, which can limit the value of the solution for certain organizations.

Anything as a Service XaaS



Hardware as
a Service



Infrastructure
as a Service



Platform as
a Service



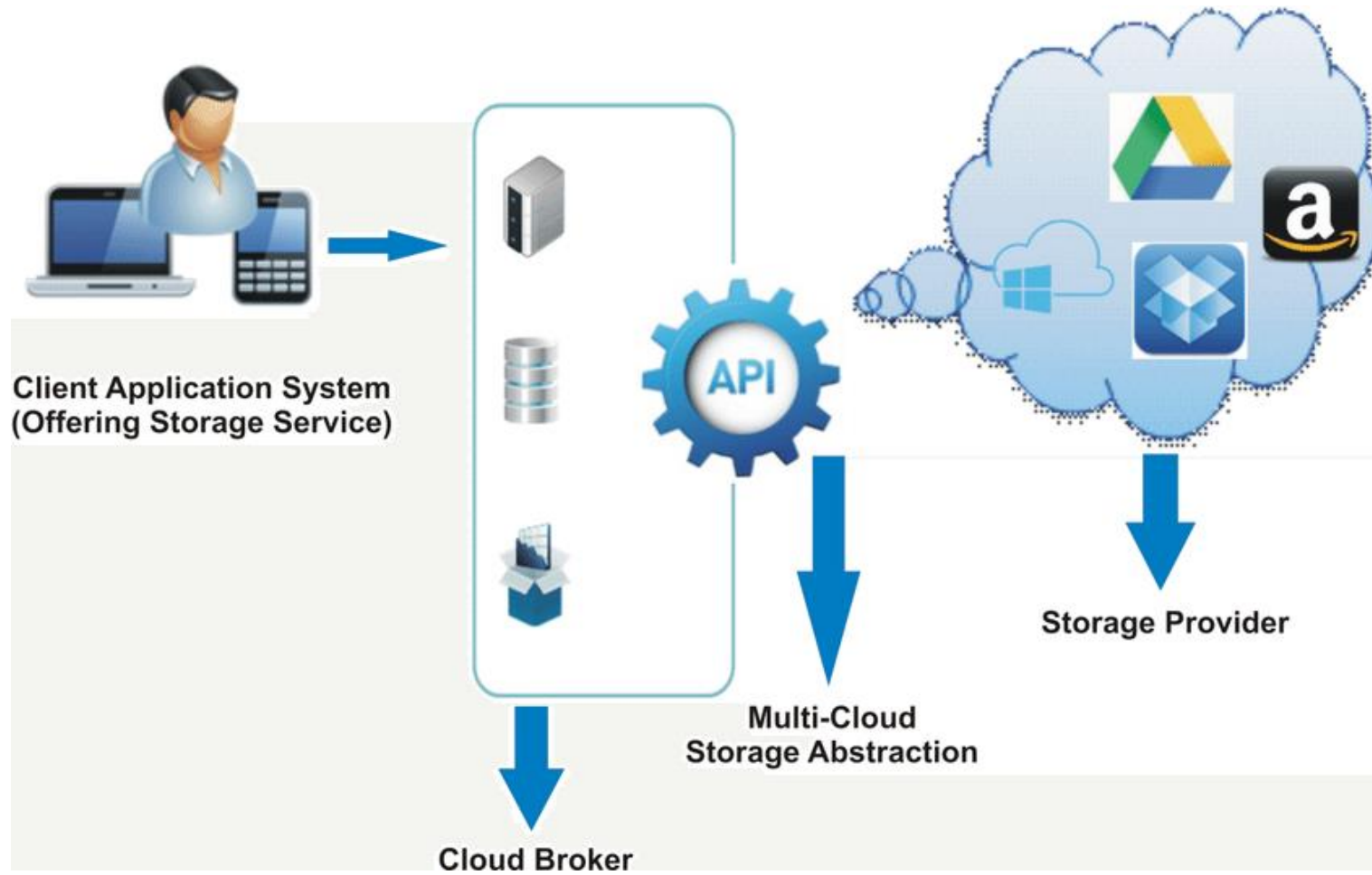
Software as
a Service

Storage as a Service (STaaS)

Today, in this data-driven world, the need for efficient and flexible storage solutions has become most important. Storage as a Service (STaaS), a cloud-based model, is the most efficient way for businesses to manage their data. It provides the ability to rent store infrastructure from providers and eliminates the burden of purchasing and maintaining their own hardware storage.

Uses of STaaS

- ▶ STaaS can be utilized by enterprises for various purposes such as data transfers, redundant storage and data recovery in case of data corruption or loss. Here we will see some more such use cases of STaaS.
- ▶ **Data Backup and Recovery:** STaaS provides better solution for data backup and recovery. Various enterprises can use it to automatically back up critical data and quickly restore it in case of data loss or corruption.
- ▶ **Entertainment and Media:** It is used in media and entertainment industry for storing and managing large media files, such as video, images and audio files. It allow then for efficient distribution and management of data.
- ▶ **Scalable Storage:** STaaS can be used by various enterprises to scale their storage infrastructure according to their needs. It helps in handling the fluctuating storage demands without investing in additional hardware.
- ▶ **Disaster Recovery:** STaaS can be used in case of any disaster recovery. It allow efficient storage for various organizations to replicate their data to remote locations for data redundancy. Which will help them in future in case of any disaster to recover their data.



Network as a Service (NaaS)

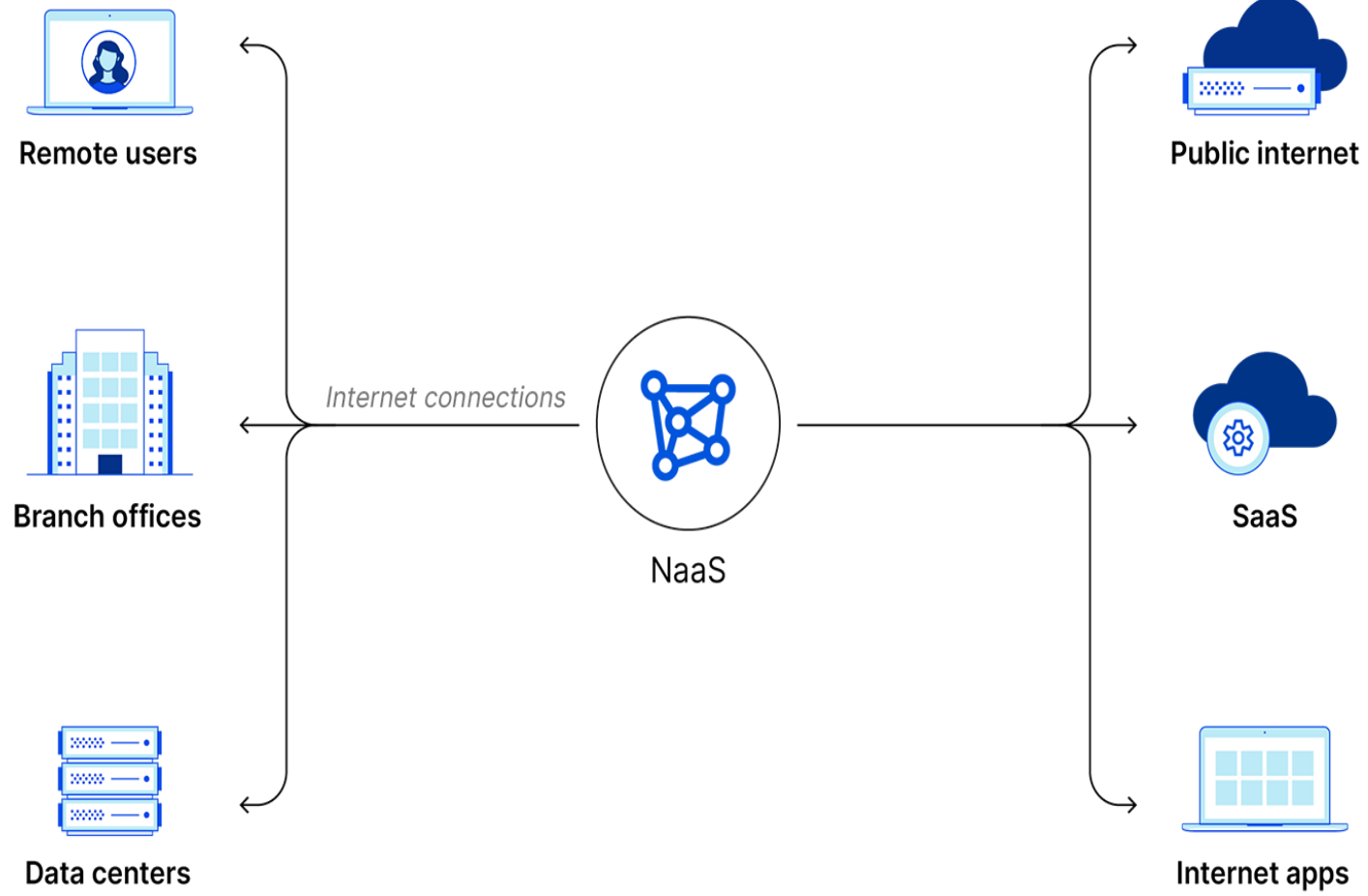
NaaS is a cloud model that enables users to easily operate the network and achieve the outcomes they expect from it without owning, building, or maintaining their own infrastructure.

NaaS can replace hardware-centric VPNs, load balancers, firewall appliances, and Multiprotocol Label Switching (MPLS) connections. Users can scale up and down as demand changes, rapidly deploy services, and eliminate hardware costs.

Advantages of NaaS

- ▶ **Flexibility:** Cloud services offer more flexibility and greater customization. Changes are made to the network via software, not hardware. IT teams are often able to reconfigure their corporate networks on demand.
- ▶ **Scalability:** Cloud services like NaaS are naturally more scalable than traditional, hardware-based services. Enterprise NaaS customers can simply purchase more capacity from a vendor instead of purchasing, plugging in, and turning on more hardware.

- ▶ **Access from anywhere:** Depending on how a cloud-based network is configured, users may be able to access it from anywhere – and on any device – without using a VPN, although this introduces the need for strong access control. Ideally, all a user needs is an Internet connection and login credentials.
- ▶ **No maintenance:** The cloud provider maintains the network, managing software and hardware upgrades.
- ▶ **Bundled with security:** NaaS makes it possible for a single provider to offer both networking services and security services like firewalls. This results in tighter integration between the network and network security.
- ▶ **Cost savings:** This advantage depends on the vendor. However, purchasing cloud services instead of building one's own services often results in cost savings: cloud customers do not need to purchase and maintain hardware, and the vendor already has the servers they need to provide the service.

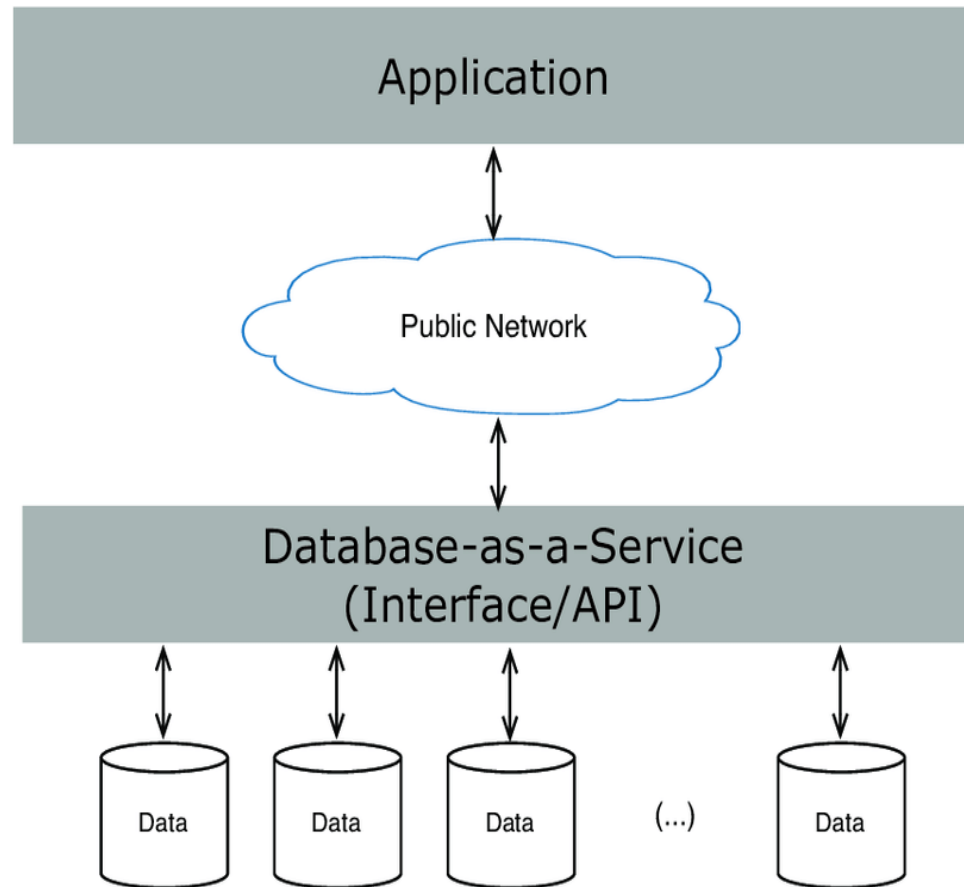


Database as a Service (DBaaS)

- ▶ Like [SaaS](#), [PaaS](#), and [IaaS](#) of cloud computing, we can consider DBaaS (also known as Managed Database Service) as a cloud computing service. It allows users associated with database activities to access and use a cloud database system without purchasing it.
- ▶ DBaaS and cloud databases come under Software as a Service (SaaS) whose demand is growing so fast
- ▶ Database as a Service (DBaaS) is self-service/on-demand database consumption coupled with automation of operations.
- ▶ As we know cloud computing services are like pay-per-use so DBaaS is also based on the same payment structure like how much you will use just pay for your usage.
- ▶ This DBaaS provides the same function as standard traditional and relational database models.
- ▶ Using DBaaS, organizations can avoid database configuration, management, degradation, and security.
- ▶ DBaaS consists of an info manager element, that controls all underlying info instances via API.
- ▶ This API is accessible to the user through a management console, typically an online application, that the user might use to manage and assemble the info and even provision or deprovision info instances.

Features of DBaaS

- ▶ **Scalability:** DBaaS platforms supports scaling resources automatically up and down based on the demand and the amount of storage and processing can be easily adjusted by the users without downtime.
- ▶ **High Availability:** In DBaaS, data is replicated over multiple servers to ensure continuous availability.
- ▶ **Automated Failover Mechanism:** Automated failover mechanism in DBaaS ensures minimal disruption in case of network or hardware failure.
- ▶ **Caching:** Frequently accessed data is cache to speed up the query response time.
- ▶ **Regular Updates:** The service provider handles regular updates, security fixes, and patches.
- ▶ **Shared Infrastructure:** Multiple tenants share the same infrastructure, thus reducing the costs.
- ▶ **Support for Multiple Database Engines:** DBaaS provides support for multiple database engines like MySQL, PostgreSQL, MongoDB, etc.
- ▶ **Cost-efficiency:** The charges are based on the actual usage, thus this is more cost-effective than the traditional database solutions.



Differences between IaaS, PaaS and SaaS

On-Premises	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking



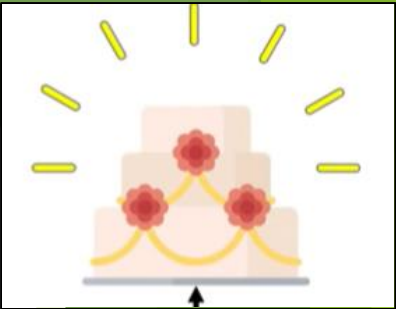
Managed by you



Managed by Vendor

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Differences between IaaS, PaaS and SaaS



On-Premises

IaaS

PaaS

SaaS

Made at Home

Buy & bake

Cake delivery

Dine out

Dinning table	Dinning table	Dinning table	Dinning table
Water	Water	Water	Water
Electricity	Electricity	Electricity	Electricity
Oven	Oven	Oven	Oven
Cake Pan	Cake Pan	Cake Pan	Cake Pan
Flour	Flour	Flour	Flour
Sugar	Sugar	Sugar	Sugar
Butter	Butter	Butter	Butter
Eggs	Eggs	Eggs	Eggs



Managed by you



Managed by Vendor

Cloud Providers



Cloud Computing with AWS



Amazon Web Services (AWS) is a cloud service from Amazon



It provides services over the internet



AWS services can be used to create and deploy any type of application in the cloud



AWS uses the subscription pricing model (pay for what you use)



Lifecycle of cloud computing solution



Define the Purpose

Understand the requirements of the business and determine what type of applications to run on the cloud



EC2



Lambda



Elastic container Service



Define the Hardware

Choose a compute service that will provide the right support where you resize the compute capacity in the cloud to run application programs



S3



EFS



Glacier



Define the Storage

Choose a storage service where you can backup and archive your data over the internet



VPC



Route 53



Direct Connect



Define the Network

Define a network that securely delivers data, videos, applications etc. with low latency and high transfer speed



CloudWatch



Auto scaling

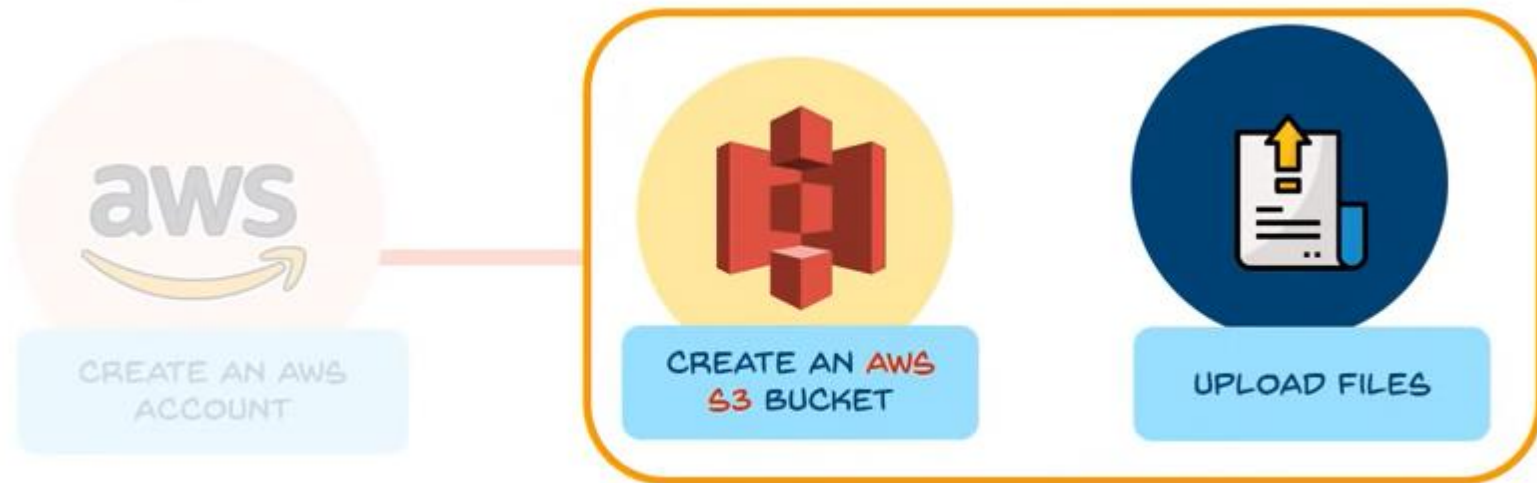


CloudFormation



Define Management Processes and Tools

You can have complete control on your cloud environment by defining management tools which monitor AWS resources and the customer applications running on AWS platform



Advantages of Cloud Computing

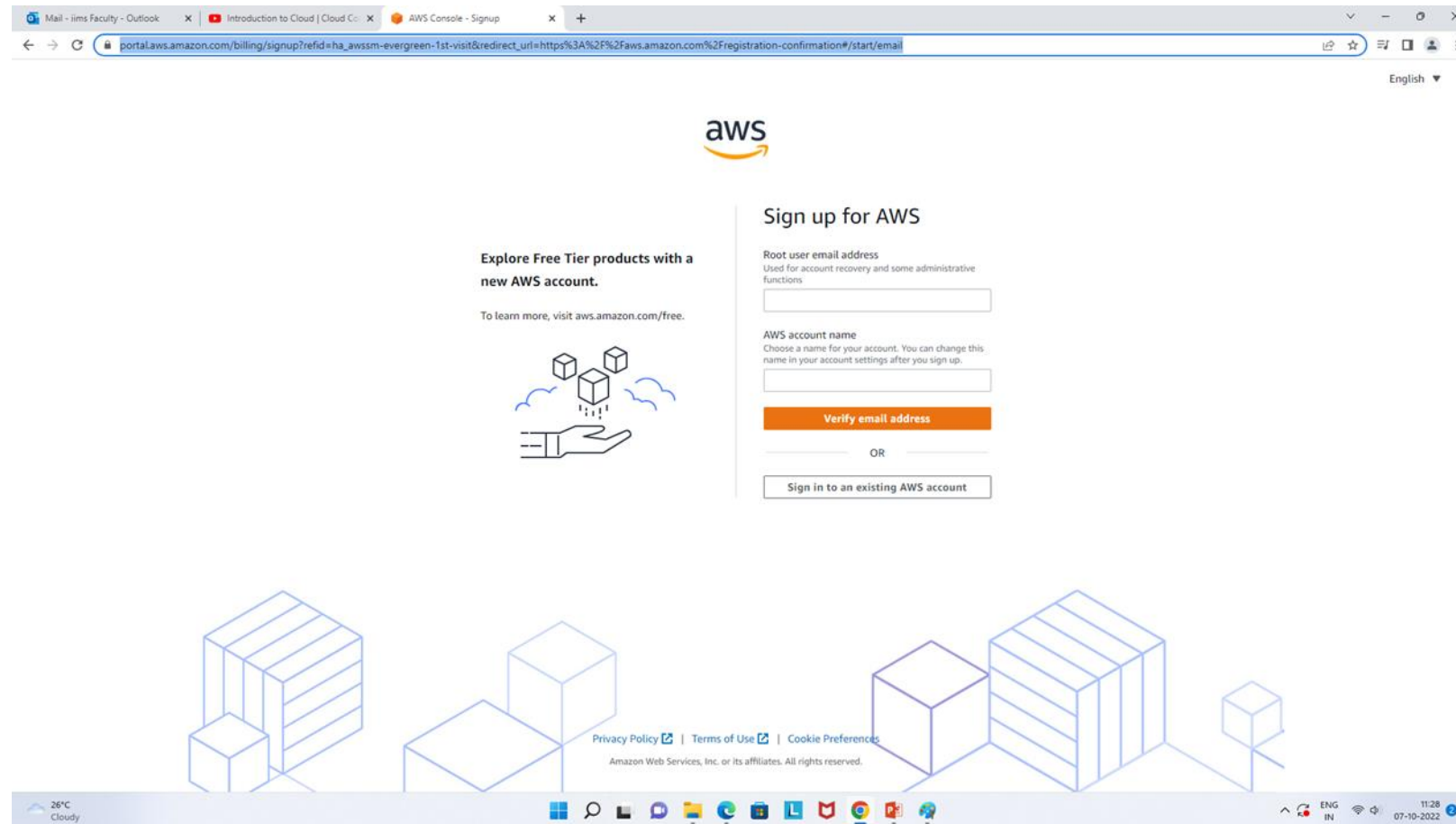
- ▶ **Scalability:** Cloud computing allows organizations to easily scale their computing resources up or down as their needs change, without having to purchase and manage additional hardware.
- ▶ **Cost Savings:** Using cloud computing can be more cost-effective than maintaining on-premises IT infrastructure, as it eliminates the need for costly hardware, software, and maintenance expenses.
- ▶ **Accessibility:** Cloud computing enables remote access to applications and data, allowing users to work from anywhere with an internet connection.
- ▶ **Reliability:** Cloud providers offer high levels of uptime and redundancy, ensuring that applications and data are available even in the event of hardware failure.
- ▶ **Flexibility:** Cloud computing offers a wide range of deployment options, including public, private, and hybrid clouds, which can be tailored to meet the unique needs of an organization.

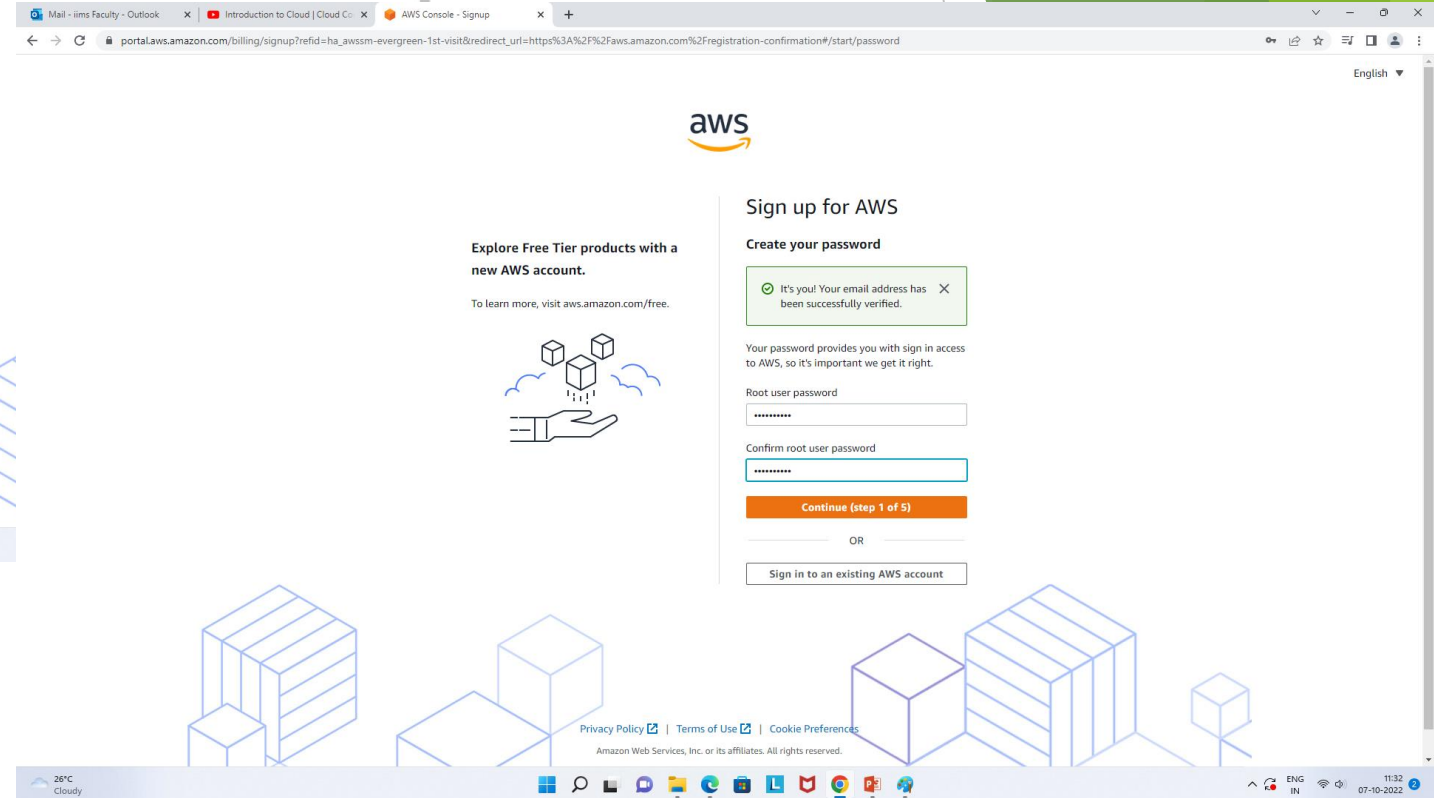
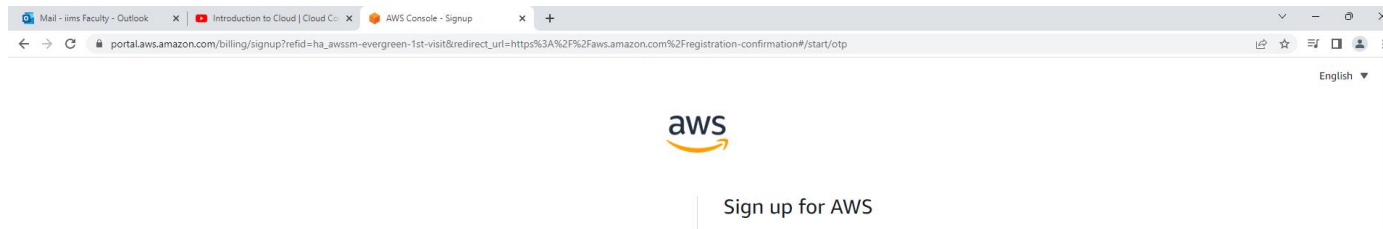
Disadvantages of Cloud Computing

- ▶ **Security:** Cloud computing involves sharing sensitive data with a third-party provider, which raises concerns about data security and privacy.
- ▶ **Dependence:** Organizations that rely on cloud providers for their computing infrastructure are vulnerable to service disruptions or data loss in the event of provider outages or other issues.
- ▶ **Internet Dependency:** Cloud computing requires a reliable and fast internet connection to access applications and data, which can be a challenge in some areas.
- ▶ **Technical Issues:** Technical issues with cloud services, such as compatibility problems or software bugs, can sometimes be difficult to resolve and can result in downtime or lost productivity.
- ▶ **Lack of Control:** With cloud computing, organizations have limited control over the infrastructure and services they use, which can make it difficult to customize or optimize the environment for their specific needs.

How to Creating cloud on AWS

- ▶ <https://aws.amazon.com/>
- ▶ https://portal.aws.amazon.com/billing/signup?refid=ha_awssm-evergreen-1st-visit&redirect_url=https%3A%2F%2Faws.amazon.com%2Fregistration-confirmation#/start/email





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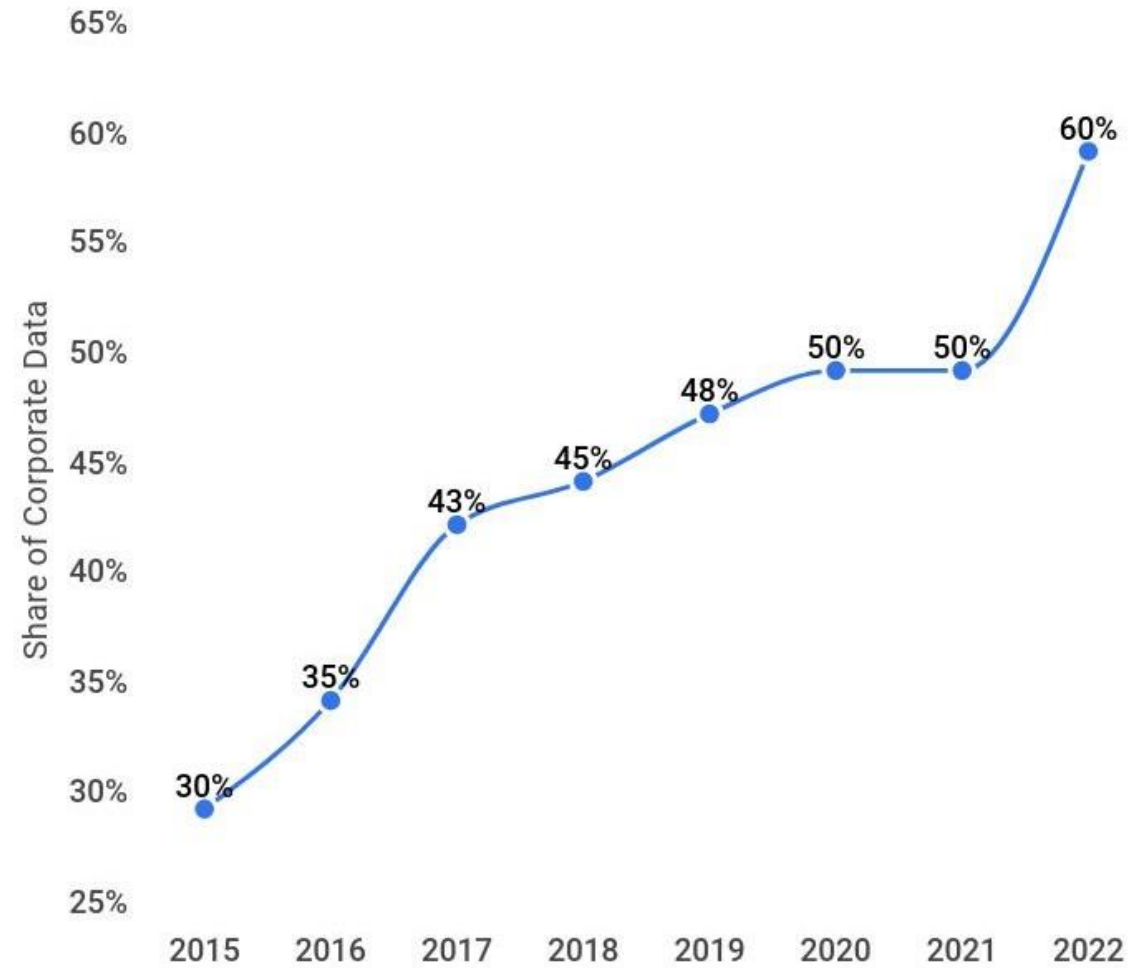
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AWS Console - Signup

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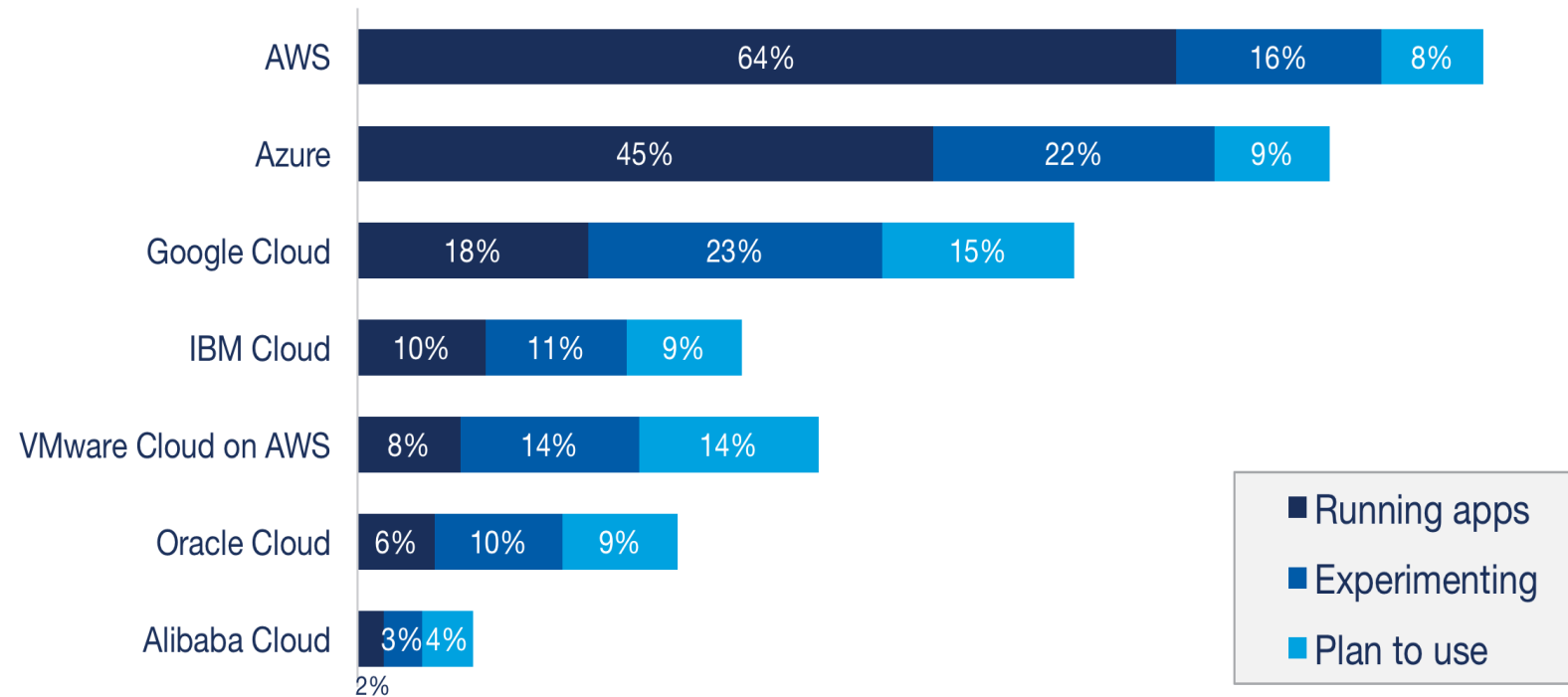
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SHARE OF CORPORATE DATA STORED IN THE CLOUD OVER TIME



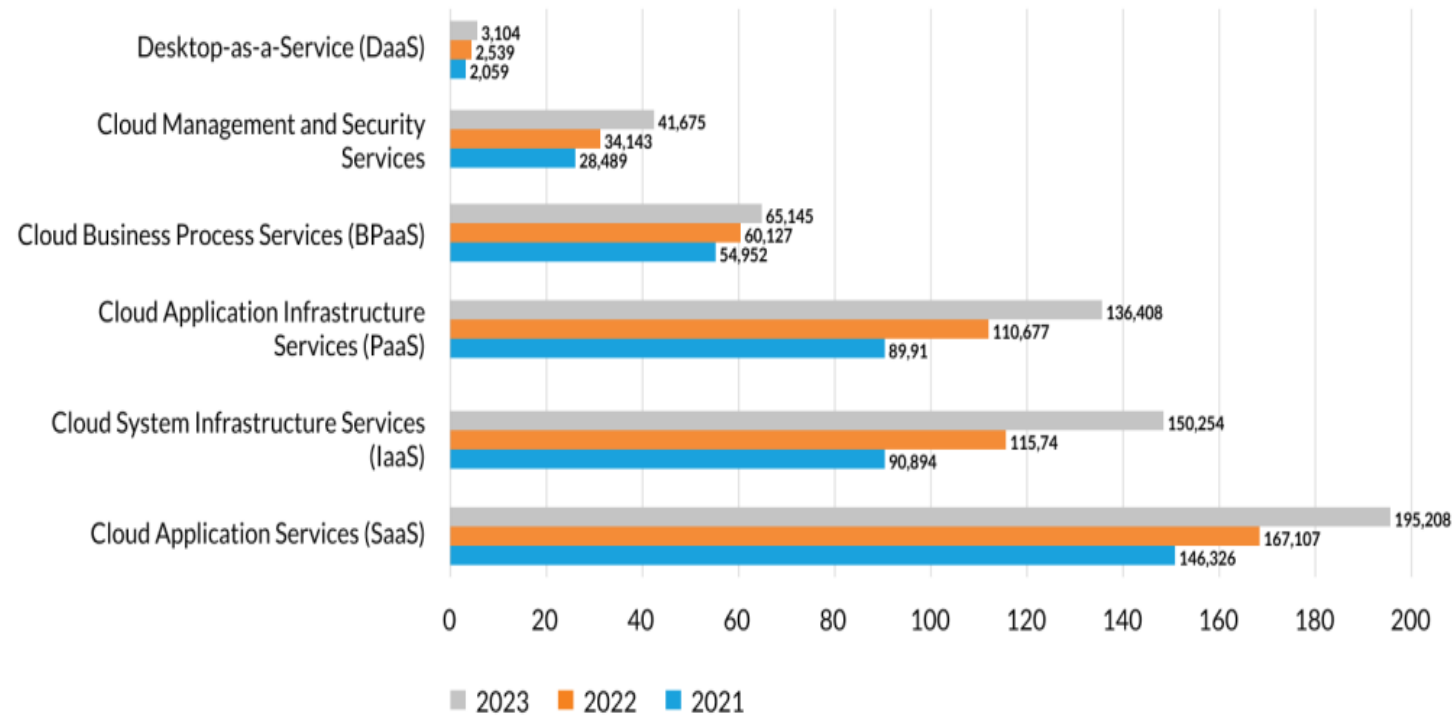
Public Cloud Adoption

% of Respondents Running Applications



Source: RightScale 2018 State of the Cloud Report

Worldwide Public Cloud Services Spending Forecast 2021-2023 (Million USD)



Source: Gartner

Design by **dp** DEV.PRO